

Self-Organization in Networking

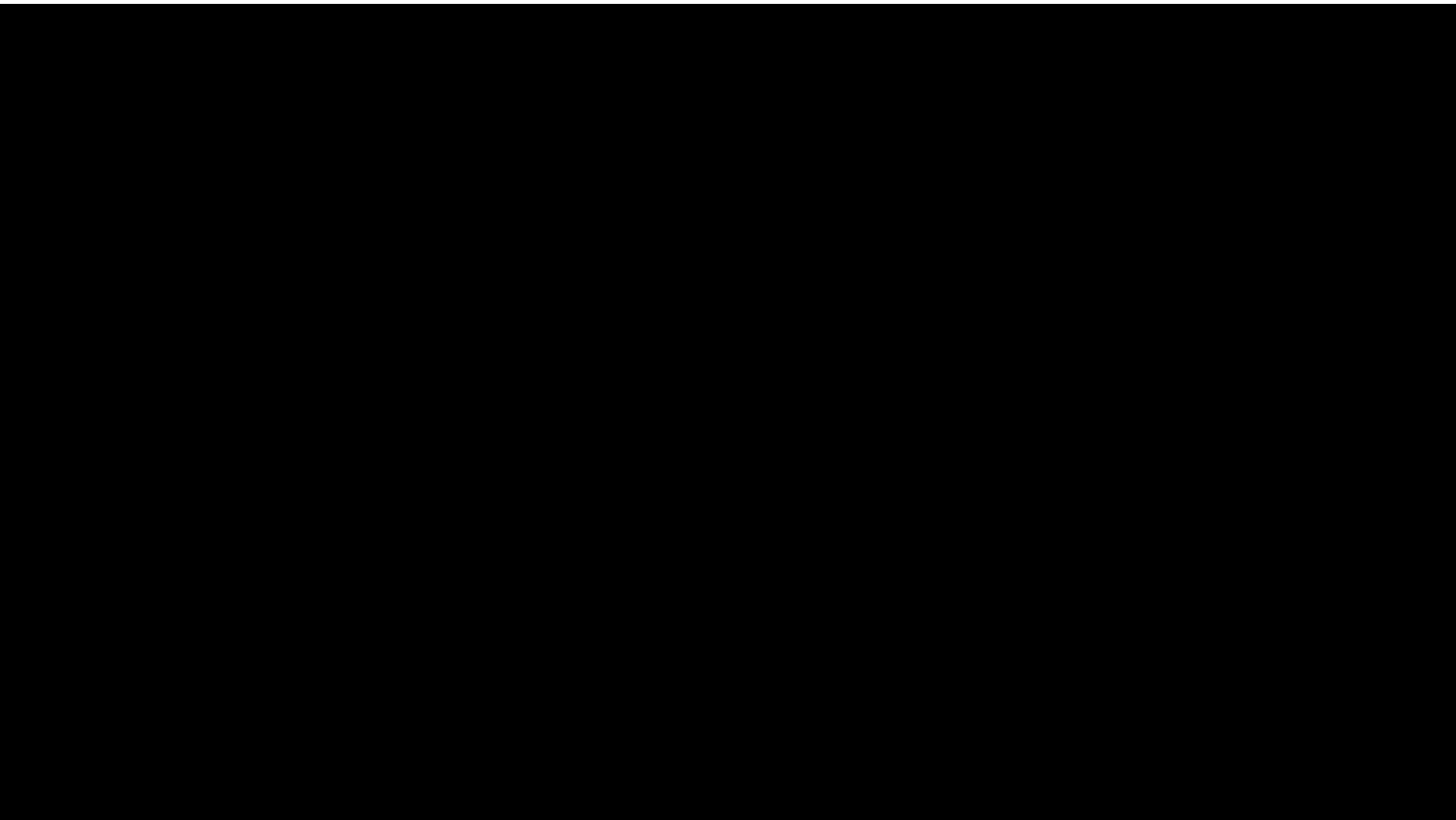
Advanced Mobile Communication Networks



Motivation for Self-Organization of Networks

- Problems of today's networks
 - Heterogeneity, dynamics, scalability
 - Manageability
 - Operational cost
- Self-organization offers
 - Autonomous optimization and repair
 - Automation of control
 - Scalability
 - Absence of single point of failure
- Some applications
 - Network management: Self-Organizing Networks (SON) in LTE
 - Protocol functions: TCP congestion control, 802.11 DCF, AODV route discovery, ...
 - Autonomous coordination of UAVs

Introduction to Self-organization



Definitions and Properties

Questions to be answered:

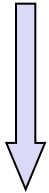
- What's the idea of self-organized (SO) systems?
- Why is SO so popular for modern systems?
- What are the properties of SO systems?
- What are the limits of SO systems?
- What does emergence mean?

From Centralized to Self-organized Systems

- Monolithic/centralized systems

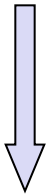
Monolithic: Systems consisting of a single computer, its peripherals, and perhaps some remote terminals

Centralized: systems with a well-defined centralized control process



- Distributed systems

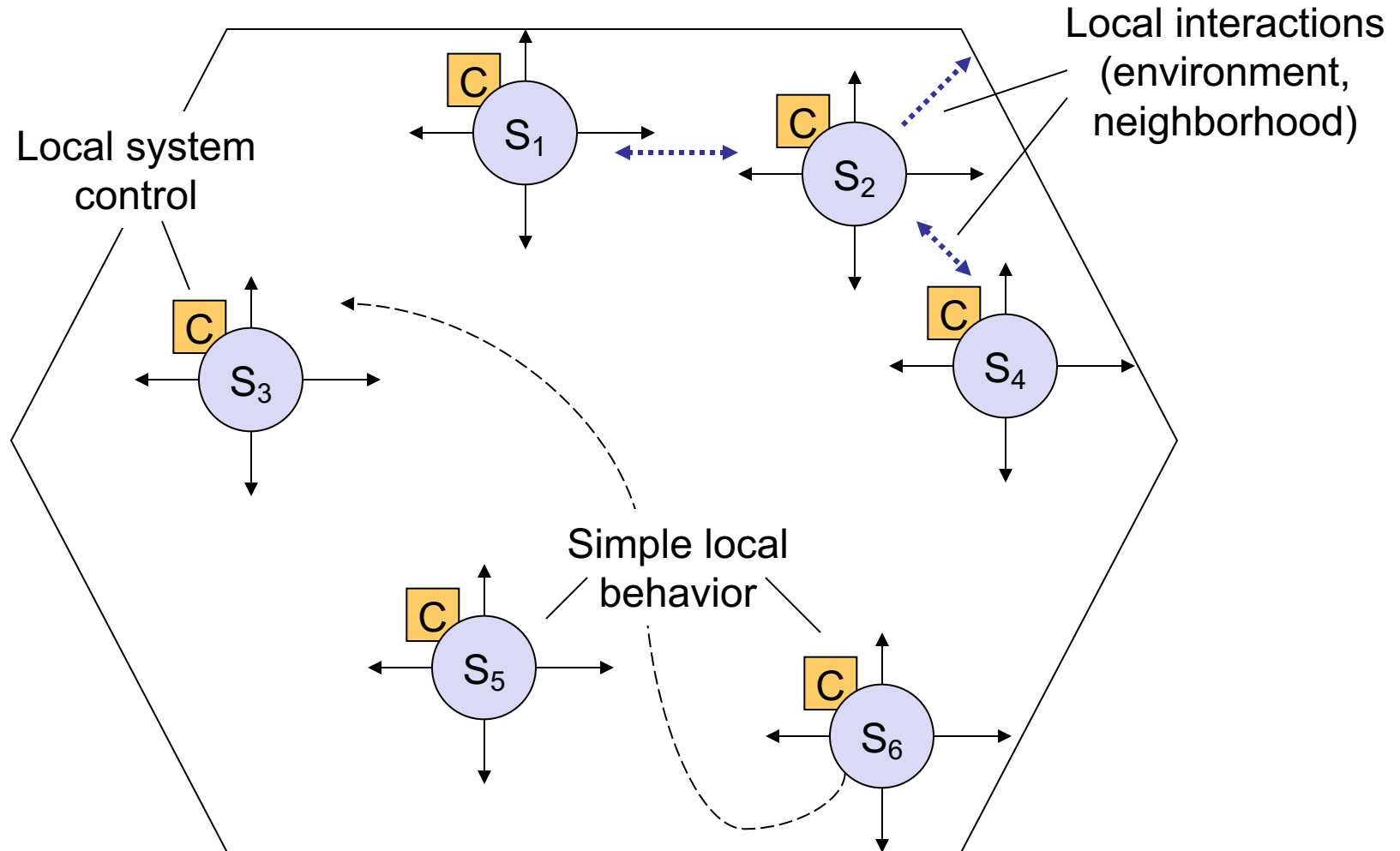
A collection of independent subsystems that appears to the application as a single coherent system



- Self-organizing autonomous systems

Autonomously acting individual systems performing local programs and acting on local data but participating on a global task, i.e. showing an emergent behavior

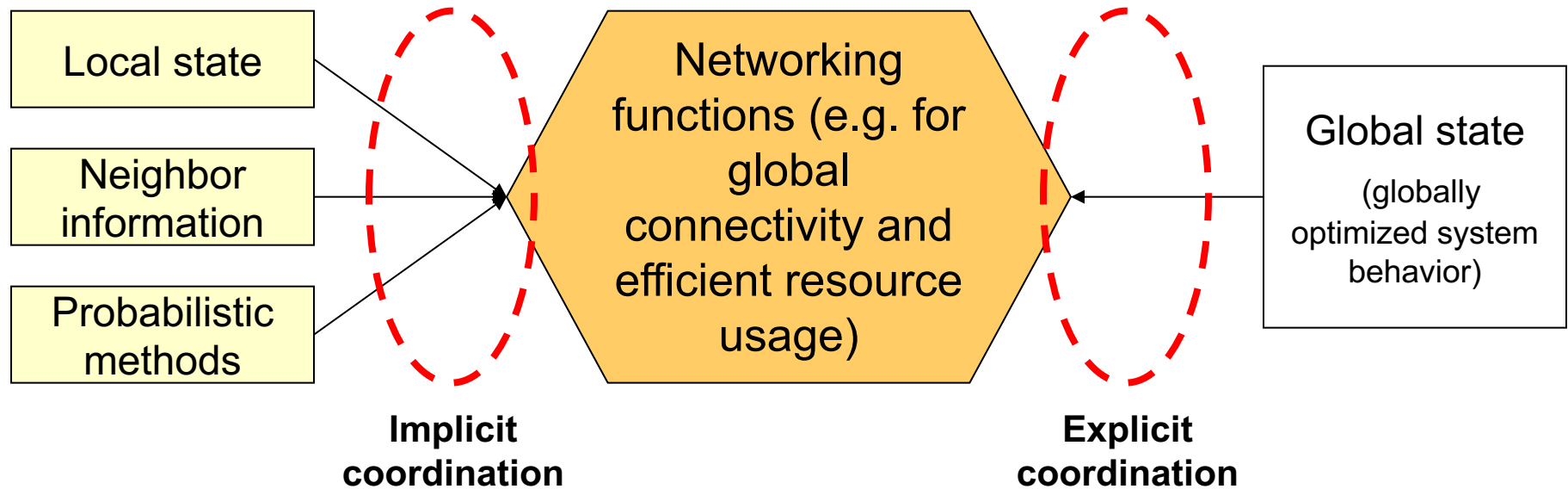
Self-organizing Systems – Operational Principles



Self-organized vs. Centralized Control

Self-organizing networks

Centralized networks



Self-organization: Definitions

Yates et al. (1987)

“**Technological systems** become organized by commands from outside, as when human intentions lead to the *building of structures* or machines.

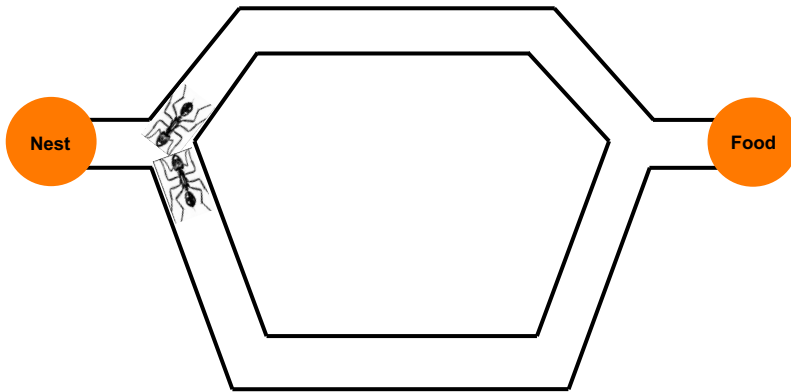
But many **natural systems** *become structured* by their own internal processes: these are the self-organizing systems, and the *emergence of order* within them is a complex phenomenon that intrigues scientists from all disciplines.”

Camazine et al. (2003)

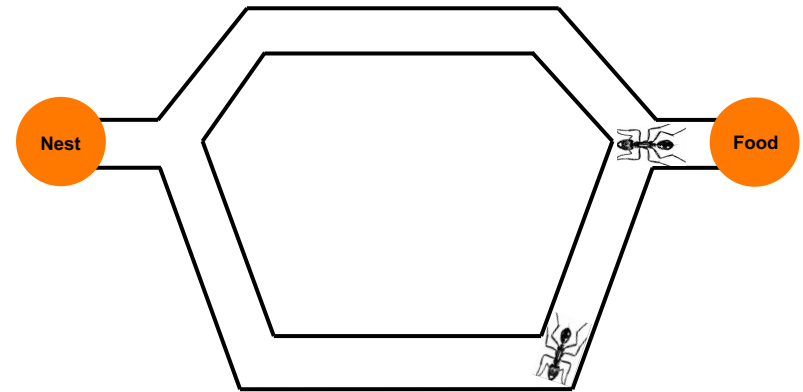
“Self-organization is a process in which ***pattern at the global level of a system emerges solely from numerous interactions among the lower-level components*** of a system.

Moreover, the rules specifying interactions among the systems' components are executed using only **local information**, without reference to the global pattern.”

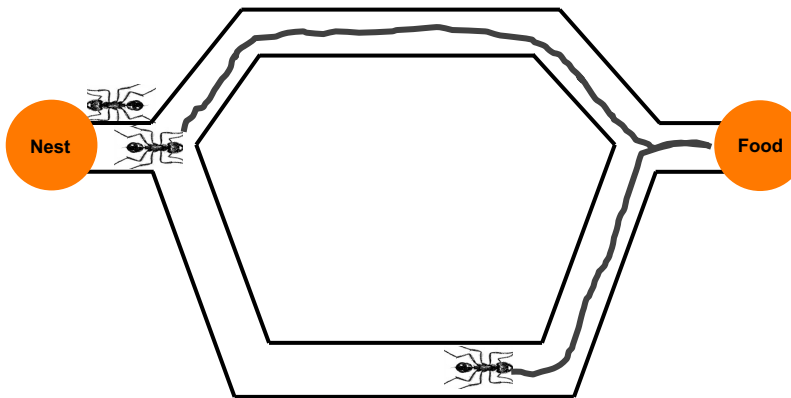
Example from Nature: Ants (Double Bridge Experiment)



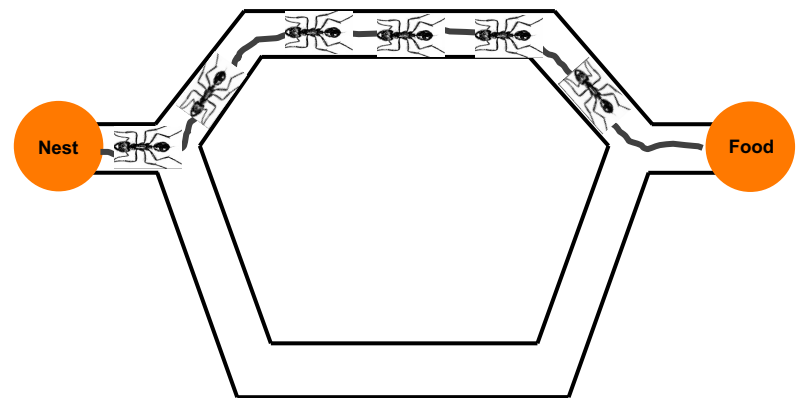
Ants start with equal probability



The ant on shorter path has a shorter time from it's nest to the food.



The density of pheromone on the shorter path is higher



After some time, the shorter path is almost exclusively used

Example from Nature: Swarm Intelligence (SI)

- A computational technique for solving distributed problems inspired from biological examples provided by
 - social insects such as ants, termites, bees, and wasps and by swarm, herd, flock, and
 - shoal phenomena such as fish shoals
- An approach for controlling and optimizing distributed systems
- Resilient, decentralized, self-organized technique



Ants



Termites



Bees



Bird flocks



Fish shoals

Swarm Intelligence – Notable Features

Autonomy:

- ➔ The system does not require outside management or maintenance; individuals are autonomous, controlling their own behavior both at the detector and effector levels in a self-organized way

Adaptability:

- ➔ Interactions between individuals can arise through direct or indirect communication

Scalability:

- ➔ SI abilities can be performed using groups consisting of a few, up to thousands of individuals with the same control architecture

Flexibility:

- ➔ No single individual of the swarm is essential, that is, any individual can be dynamically added, removed, or replaced

Robustness:

- ➔ No central coordination takes place, which means that there is no single point of failure

Massively parallel:

- ➔ Tasks performed by each individual within its group are the same

Self-organization:

- ➔ The intelligence exhibited is not present in the individuals, but rather emerges somehow out of the entire swarm

Example from Chemistry

Pattern formation in the Belousov-Zhabotinski reaction



Photography by Juraj Lipscher

Self-organization – More Examples

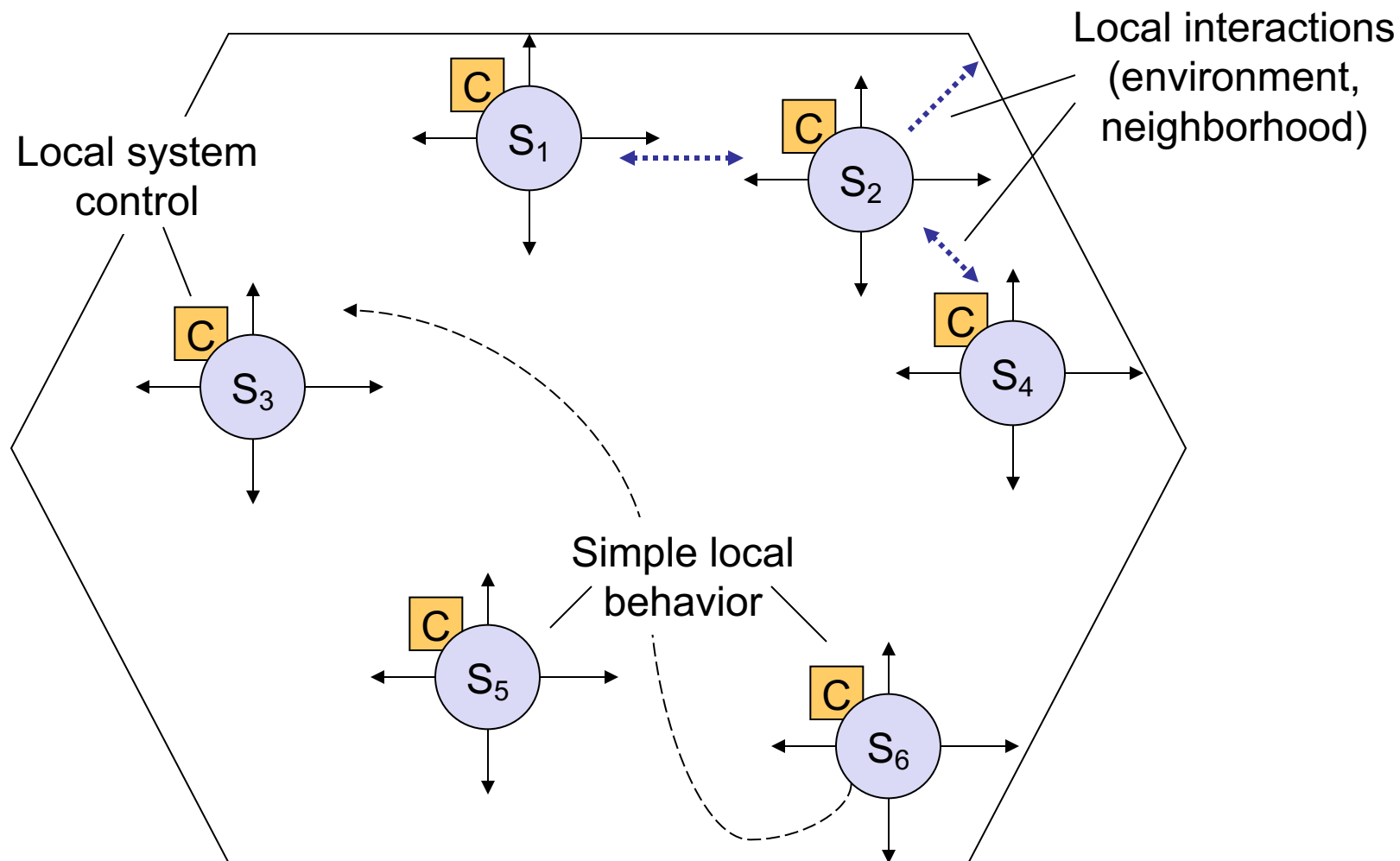
Examples of self-organized systems:

- **biology**: ants building a ant colony, a flock of birds flying in some joint and coordinated way
- **economy**: capitalistic economy with minimal rules and lots of individuals with ideas and goals representing local algorithms
- **human society**: social networks, new political forces, communities, associations, etc.

But: From time to time there may be unexpected breakdowns or crashes of the system, e.g.

- **human societies**: war, civil war
- **economy**: crash of financial markets
- **nature**: natural catastrophes

Self-organizing Systems – Operational Principles



Self-organization – System Properties

Property	Description
No central control	No global control system No central information, information is distributed Subsystems perform completely autonomous
Emerging structures	Global behavior or functioning of the system emerges in form of observable pattern or structures
High resulting complexity	Even if the individual subsystems can be simple as well as their basic rules, the resulting overall system becomes complex and often unpredictable
High scalability	No performance degradation if more subsystems are added to the system System performs as requested regardless of the number of subsystems

Self-organization – System Properties

- Absence of external control
- Global order and local interactions
- Fluctuations and instability
- Dissipation (waste of resources)
- Multiple equilibria and local optima
- Adaptation to changing conditions
- Redundancy
- Self-maintenance
- Complexity of understanding

Properties: Self-X Capabilities

Feature	Description
Self-configuration	Methods for (re-)generating adequate configurations depending on the current situation in terms of environmental circumstances
Self-management	Capability to maintain systems and devices depending on the current system parameters
Self-adaptability	Ability of the system's components to adapt to changing environmental conditions
Self-diagnosis	Mechanisms to perform autonomous checks and to compare the results with reference values
Self-protection	Capability to protect the system and its components against unwanted or even aggressive environmental influences
Self-healing	Methods for changing configurations and operational parameters of the overall system to compensate failures
Self-repair	Similar to self-healing but focusing on actual repair mechanisms for failing system parts
Self-optimization	Ability of the system to optimize the local operation parameters according to global objectives

Properties: Self-organization and Emergence

Definition ***Self-Organization*** (Camazine rephrased)

Self-organization is a process in which **structure and functionality (pattern) at the global level of a system emerge solely from numerous interactions among the lower-level components of a system** without any external or centralized control. The system's components interact in a local context either by means of direct communication or environmental observations **without reference to the global pattern.**

Definition ***Emergence***

Emergent behavior of a system is provided by the apparently meaningful collaboration of components (individuals) in order to **show capabilities of the overall system (far) beyond the capabilities of the individual components.**

Characteristics of Self-organized Systems

Self-organizing systems are dynamic and exhibit emergent properties

Since these system-level properties arise unexpectedly from nonlinear interactions among a system's components, the term emergent property may suggest to some a mysterious property that materializes magically.

Example: growth rate of a population

$$x_{n+1} = rx_n(1 - x_n)$$

$0 < r < 1$: extinction

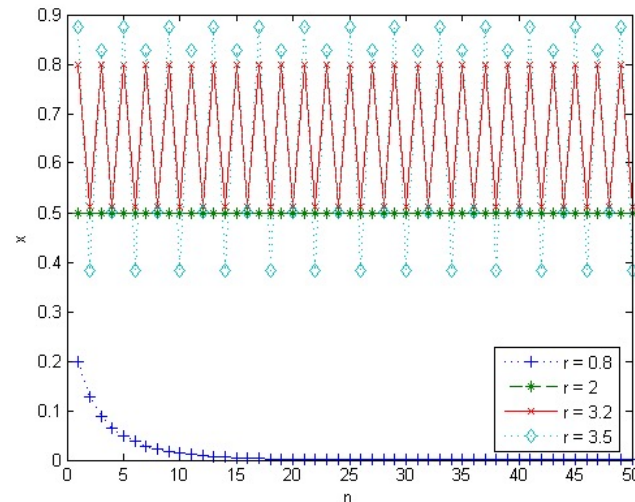
$1 < r < 3$: constant size after several generations

$3 < r < 3.4$: oscillating between two values

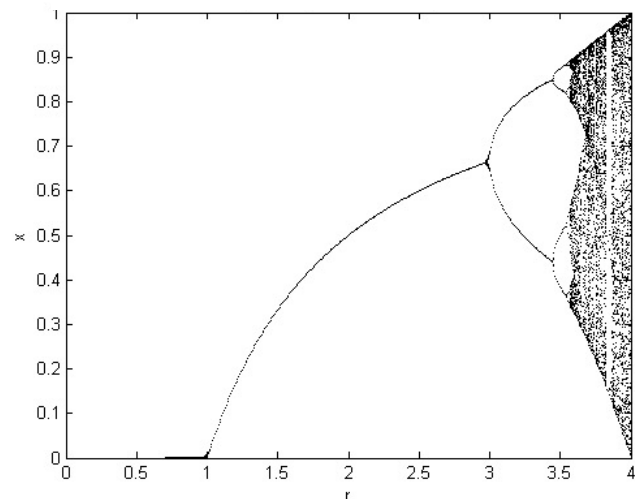
$3.4 < r < 3.57$: oscillating between four values

$r > 3.57$: "deterministic" chaos

State over time



Convergence/bifurcation depending on r



Consequences of Emergent Properties

A small change in a system parameter can result in a large change in the overall behavior of the system

- Adaptability
- Flexibility

Role of environmental factors

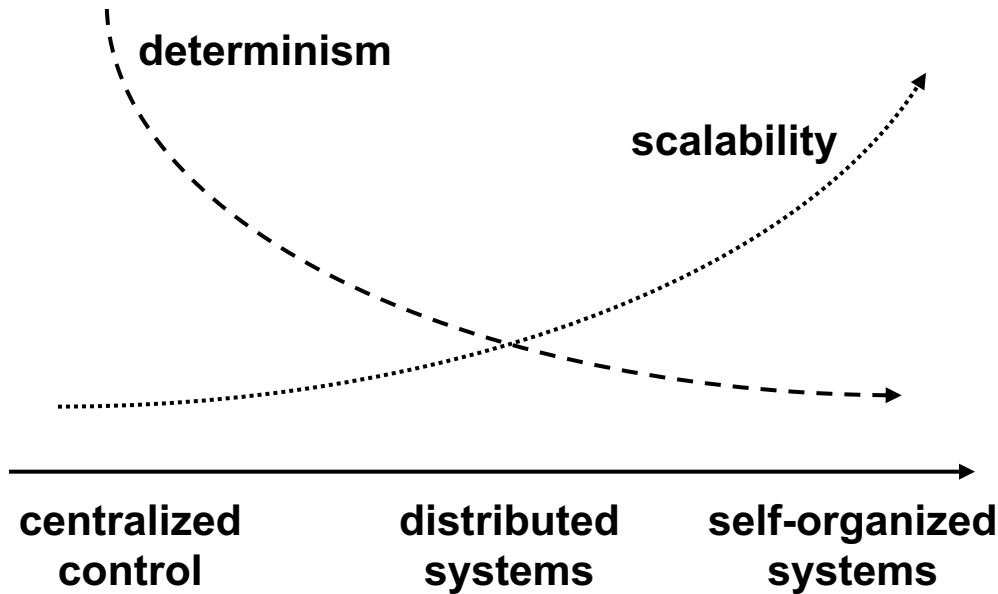
- Specify some of the initial conditions
- Positive feedback results in great sensitivity to these conditions

Self-organization and the evolution of patterns and structure

- Intuitively: generation of adaptive structures and patterns by tuning system parameters in self-organized systems rather than by developing new mechanisms for each new structure
- However: the concept of self-organization alerts us to the possibility that strikingly *different patterns* result from the *same mechanisms* operating in a different parameter range

Simple rules, complex patterns – the solution to a paradox?

Challenges and Limitations of Self-organization



- Loss of determinism
- High effort for evaluation and testing
- Resulting in high effort for development

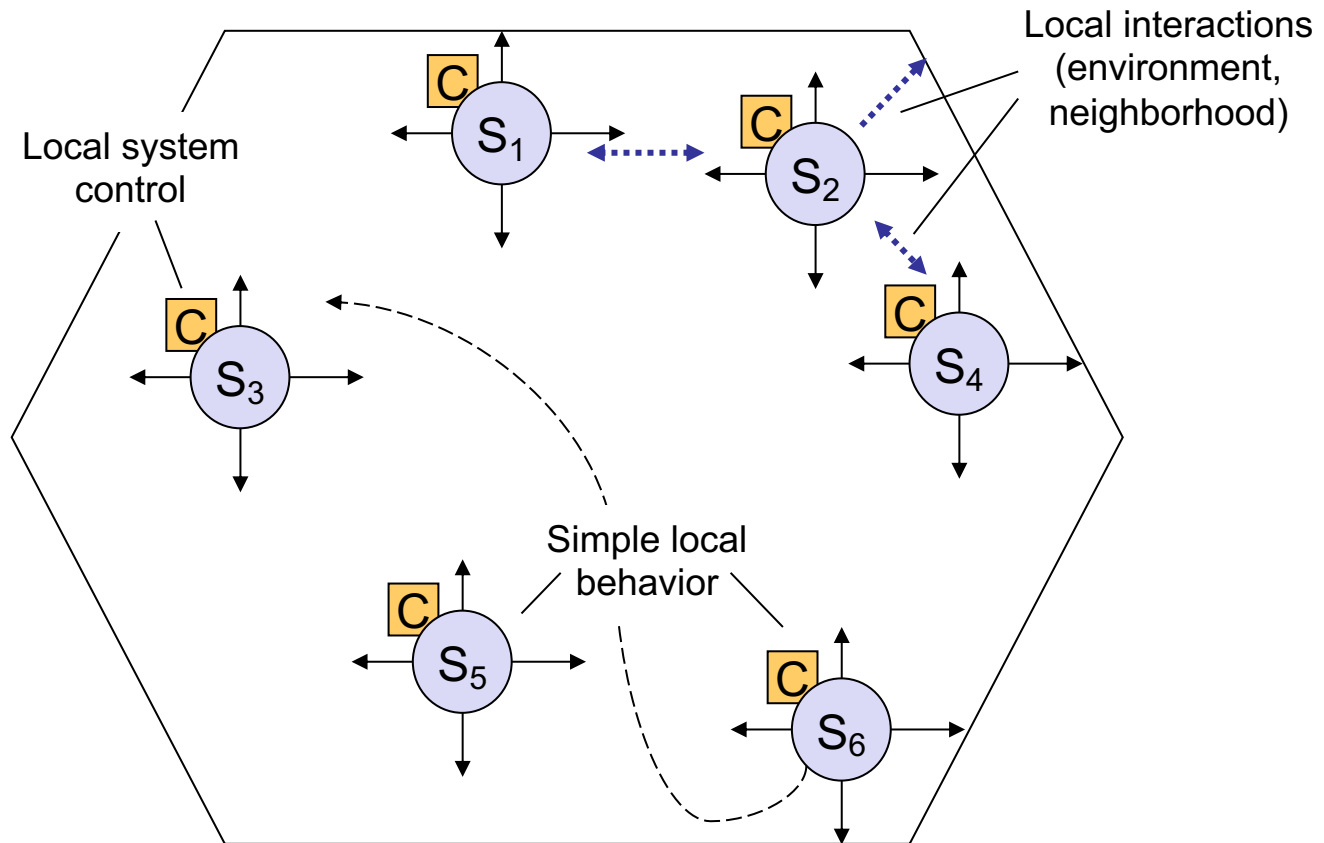
Design of Self-organized Systems

Questions to be answered:

- Identify important design principles of self-organized systems
- Identify the design paradigms for algorithms of self-organized systems
- How are these paradigms applied to communication protocols?

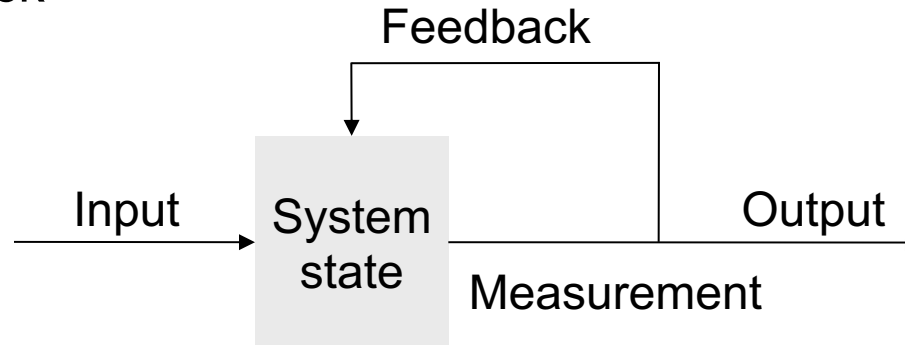
Design of Self-organized Systems

- Positive and negative feedback
- Interactions among neighbors and with the environment
- Probabilistic techniques

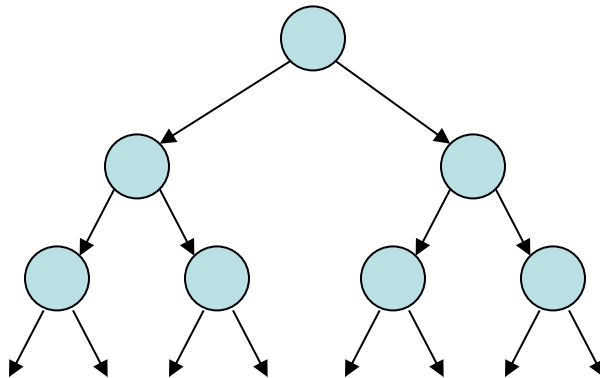


Design: Positive and Negative Feedback

Simple feedback

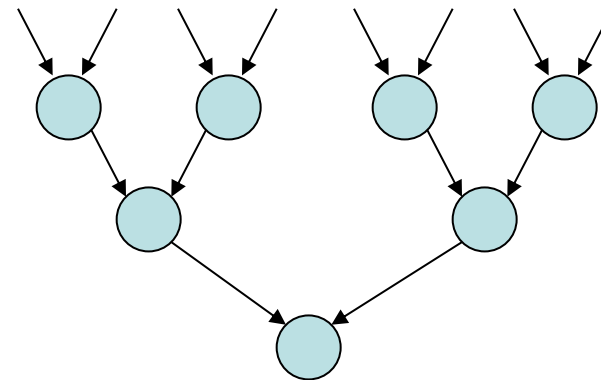


Amplification problems



Snowballing effect

(a single action or event somewhere in the system generates a global phenomenon or problem)

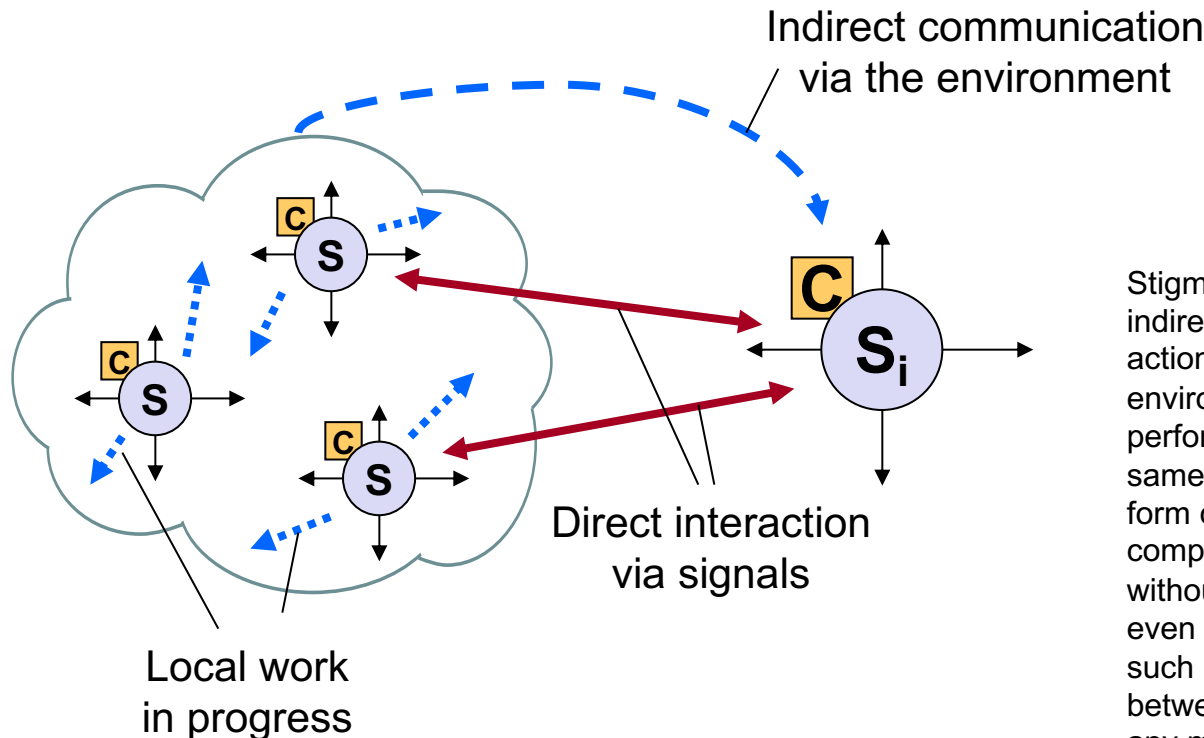


Implosion effect

(multiple actions taken simultaneously in the global system result in some blowup or disaster somewhere in the system)

Design: Interactions among Neighbors and Environment

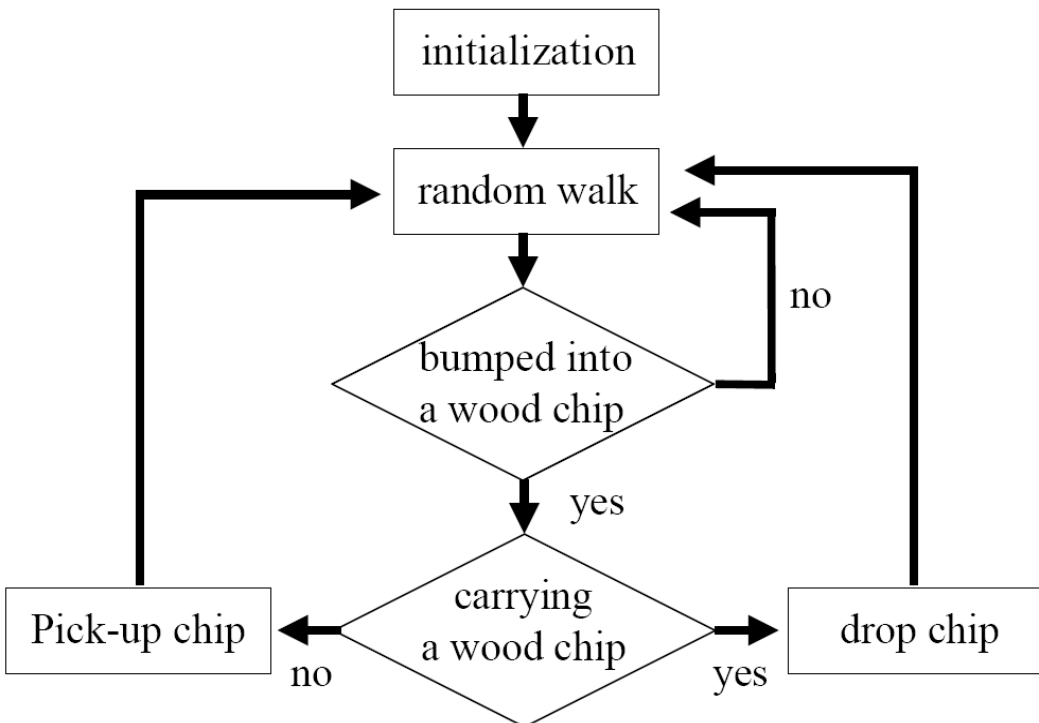
- Direct communication among neighboring systems
- Indirect communication via the environment (stigmergy)
- Interaction with (stimulation by) the environment



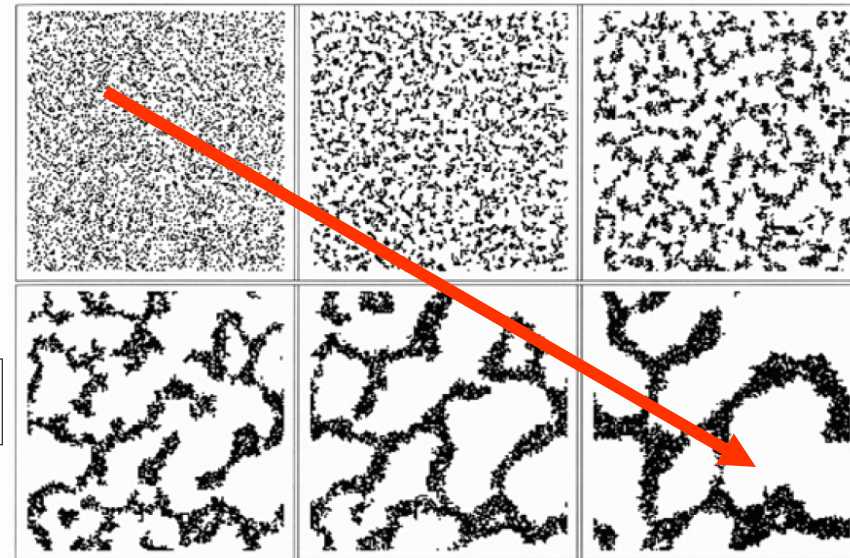
Stigmergy is a mechanism of spontaneous, indirect coordination between agents or actions, where the trace left in the environment by an action stimulates the performance of a subsequent action, by the same or a different agent. Stigmergy is a form of self-organization. It produces complex, apparently intelligent structures, without need for any planning, control, or even communication between the agents. As such it supports efficient collaboration between extremely simple agents, who lack any memory, intelligence or even awareness of each other.

Design: Probabilistic Techniques

- Examples: stochastic processes, random walk
- Objectives: leaving local optima, stabilization



Simulation results



Design Paradigms for Self-organized Systems

Paradigm #1:

Design local behavior rules that achieve global properties

Paradigm #2:

Do not aim for perfect coordination: exploit implicit coordination

Paradigm #3:

Minimize long-lived state information

Paradigm #4:

Design protocols that adapt to changes

Design Paradigms for Self-organized Systems

Paradigm #1: Design local behavior rules that achieve global properties

To design a network function that establishes a global property, this paradigm is to distribute the responsibility among the individual entities: **no single entity is "in charge" of the overall organization, but each contributes to a collective behavior.** Following this paradigm, localized behavior rules must be devised, if applied in all entities, automatically lead to the desired global property (or at least approximate it). "Localized" means that **entities have only a local view of the network** and interact with their neighbors and changes in the network have initially only local consequences.

Paradigm #2: Do not aim for perfect coordination: exploit implicit coordination

Implicit coordination means that **coordination information is not communicated explicitly by signaling messages**, but is inferred from the local environment. A node observes other nodes in its neighborhood; based on these observations, it draws conclusions about the status of the network and reacts accordingly.

Paradigm #3: Minimize long-lived state information

To achieve a higher level of self-organization, the amount of long-lived state information should be minimized. In general, the more localized the interactions are and the less coordination is needed, the less state information must be maintained.

Design Paradigms for Self-organized Systems

Paradigm #4: Design protocols that adapt to changes

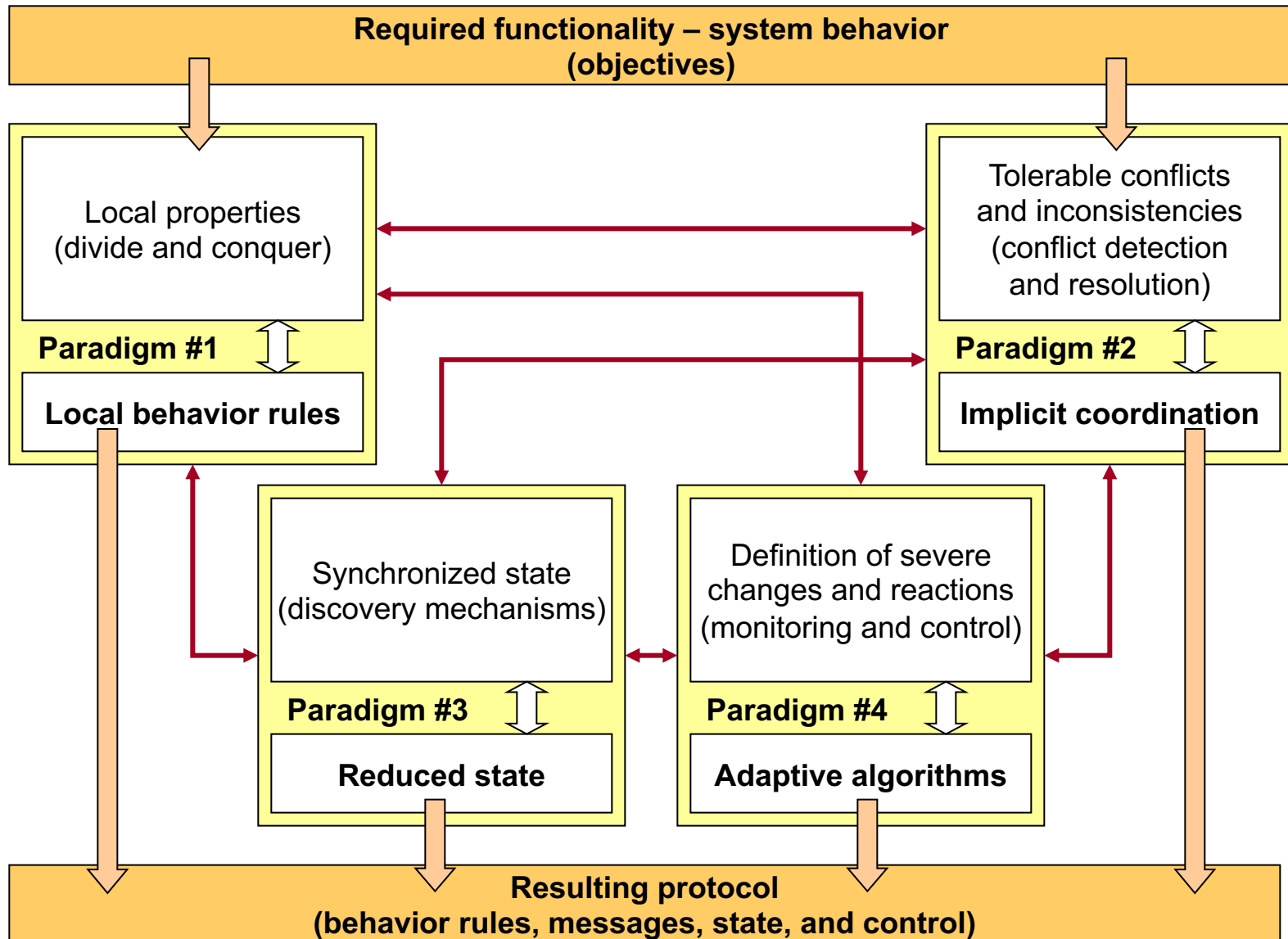
The need for the capability of nodes to react to changes in the network and its environment typically arises from changed resource constraints, changed user requirements, node mobility, or node failures. Since there are no centralized entities that could notify the nodes about changes, each node has to continuously monitor its local environment and react in an appropriate manner. Three levels of adaptation can be distinguished:

- Level 1: A protocol is designed so that it can **cope with changes**, such as failure and mobility.
- Level 2: A protocol is designed to **adapt its own parameters** (e.g., value of timers, cluster size) as a reaction to changes in order to optimize system performance.
- Level 3: A protocol is designed so that it **realizes if the changes are so severe that the currently employed mechanism is no longer suitable**. To detect such situations the environment is observed, and significant changes in major parameters trigger a fallback or alternative solution.

Typically, these levels of adaptation are combined using control loops.

Source: Christian Prehofer, Christian Bettstetter: Self-Organization in Communication Networks: Principles and Design Paradigms. IEEE Communications Magazine, July 2005

Design Paradigms for Self-organized Systems



Applications of Self-organization to Networking

Questions to be answered:

- Why are the different examples considered examples of SO? Which aspect exactly is considered self-organizing
- Identify how feedback is implemented in the different applications of SO; identify positive and negative feedback separately
- Identify the implementation of communication in the example applications
- Identify how stochastic aspects are integrated each of the examples
- Identify which level of the design paradigm 4 is implemented by the different examples

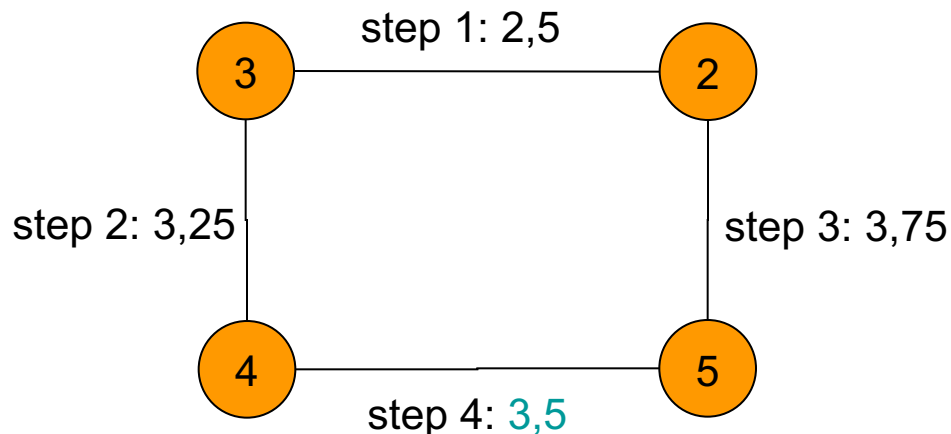
Applications of Self-organization to Networking

- Media Access Control:
 - CSMA/CD (802.3/Ethernet)
 - 802.11 DCF
 - Cognitive radio networks
- Load/congestion control:
 - Exponential backoff (CSMA systems)
 - TCP congestion control
- Routing: AODV route discovery and maintenance
- Election: cluster head selection in sensor networks (LEACH)
- Energy management: 802.11 ad hoc mode
- Network management:
 - LTE overload control
 - LTE energy saving
 - LTE mobility robustness optimization (HO parameter optimization)
 - Inter-system load balancing
- ...

Example: Information Management

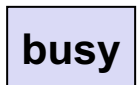
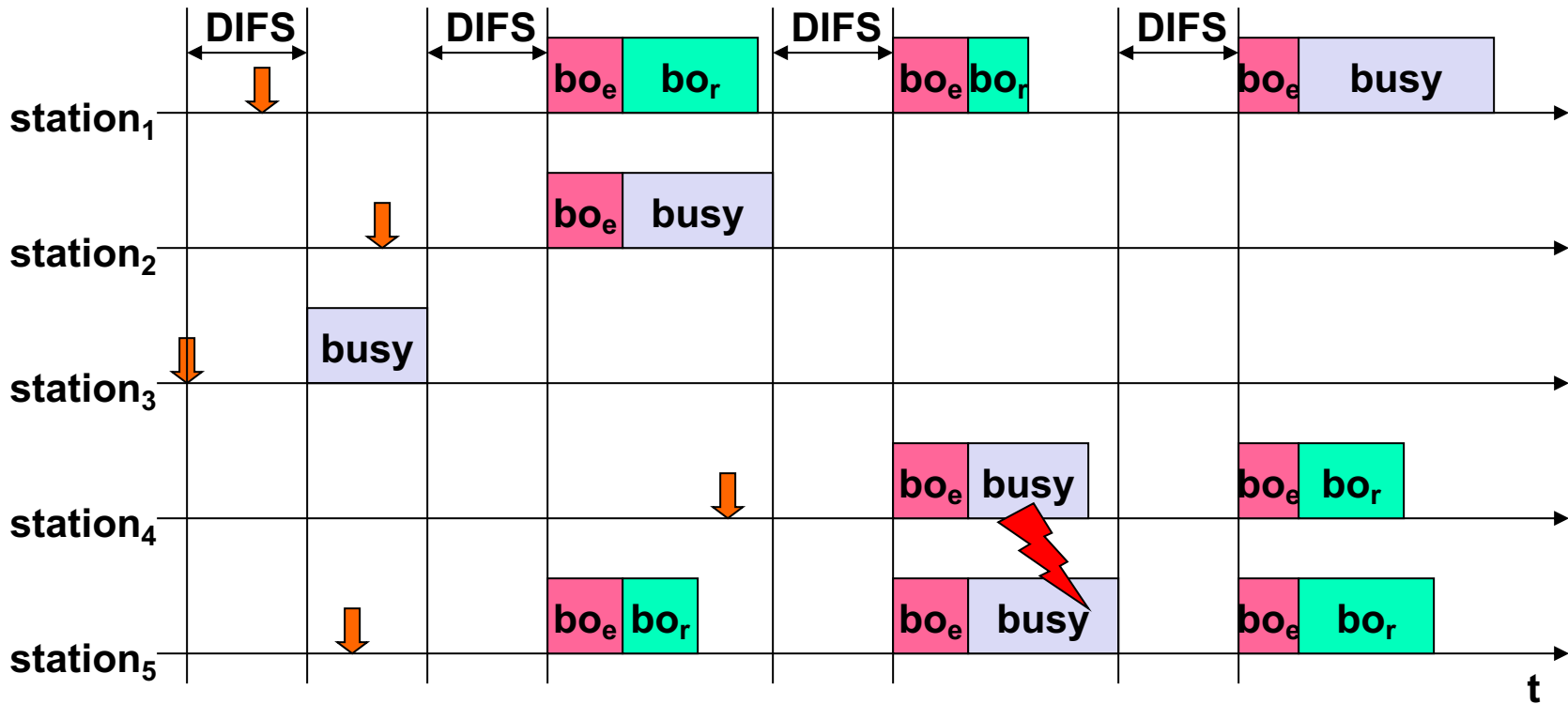
Gossip-based Aggregation

- Compute a (global) aggregate value from distributed data
 - load balancing in networks
 - monitoring in industrial control applications
- Each node
 - maintains a local approximate value of the aggregate
 - communicates periodically with randomly chosen neighbors and updates the approximate value
- Only local pairwise interaction!



Example: 802.11 DCF MAC

Distributed Coordination Function (DCF) with Exponential Backoff



medium not idle (frame, ack etc.)



elapsed backoff time



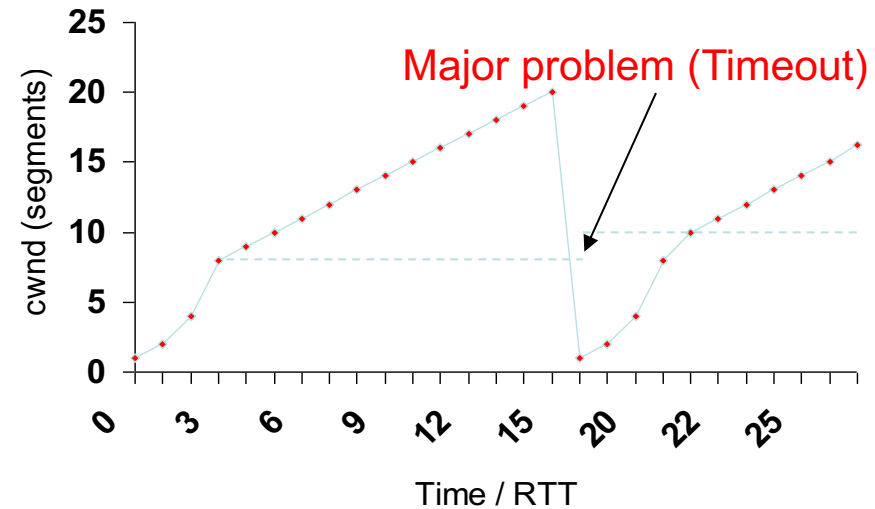
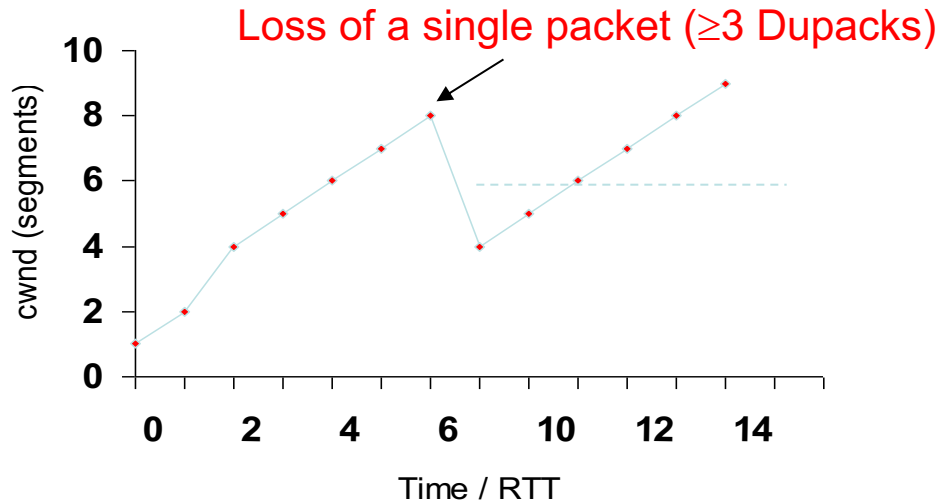
packet arrival at MAC



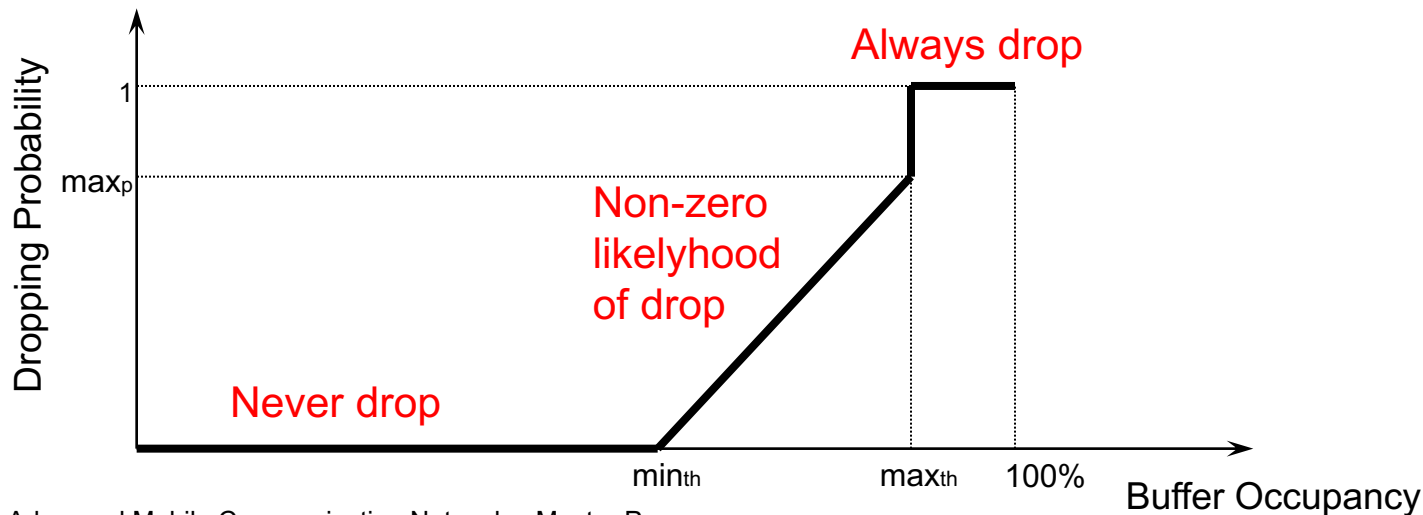
residual backoff time

Example: TCP Congestion Control and RED

TCP flow control



Random Early Detection (RED)

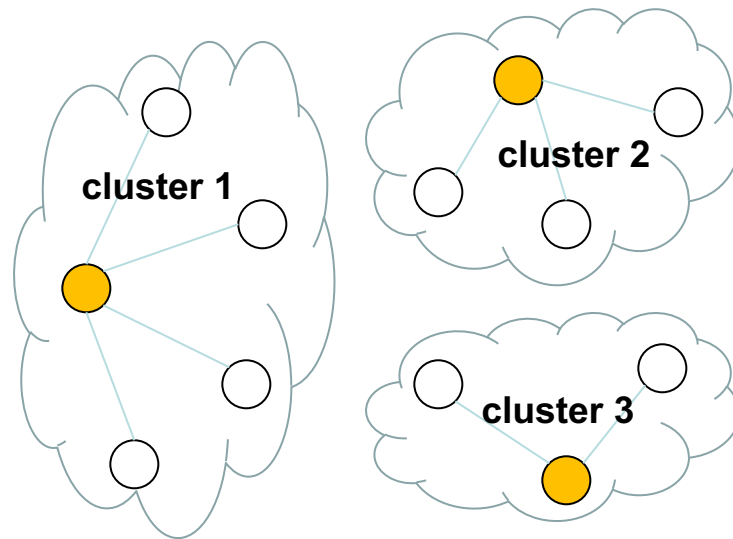


Example: Cluster Head Election in Sensor Networks

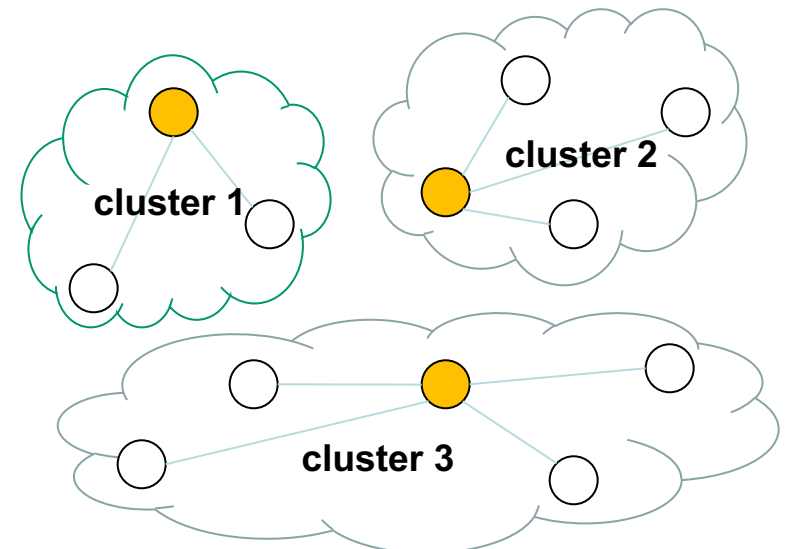
LEACH: Low-Energy Adaptive Clustering Hierarchy

Principles

- nodes elect themselves to become cluster heads at any given time with a certain probability (in HEED extension based on remaining energy)
- the cluster heads broadcast their status to the other nodes in the network
- each node determines to which cluster it wants to belong by choosing the closest cluster head (e.g. which requires the minimum communication energy)

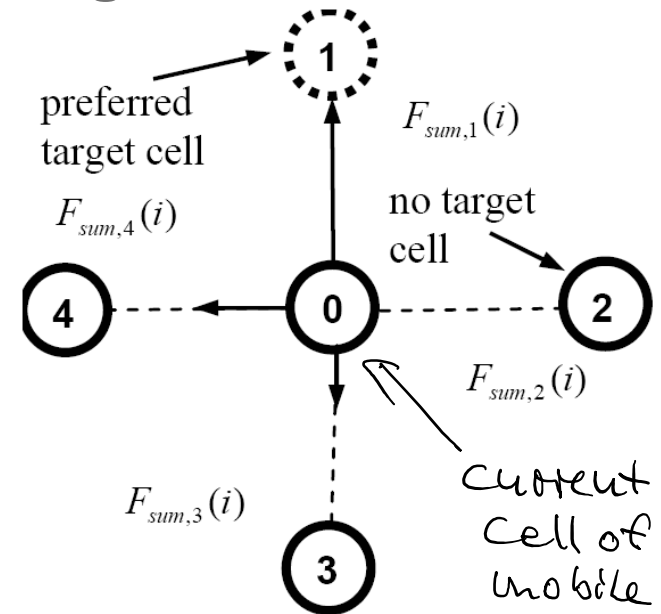
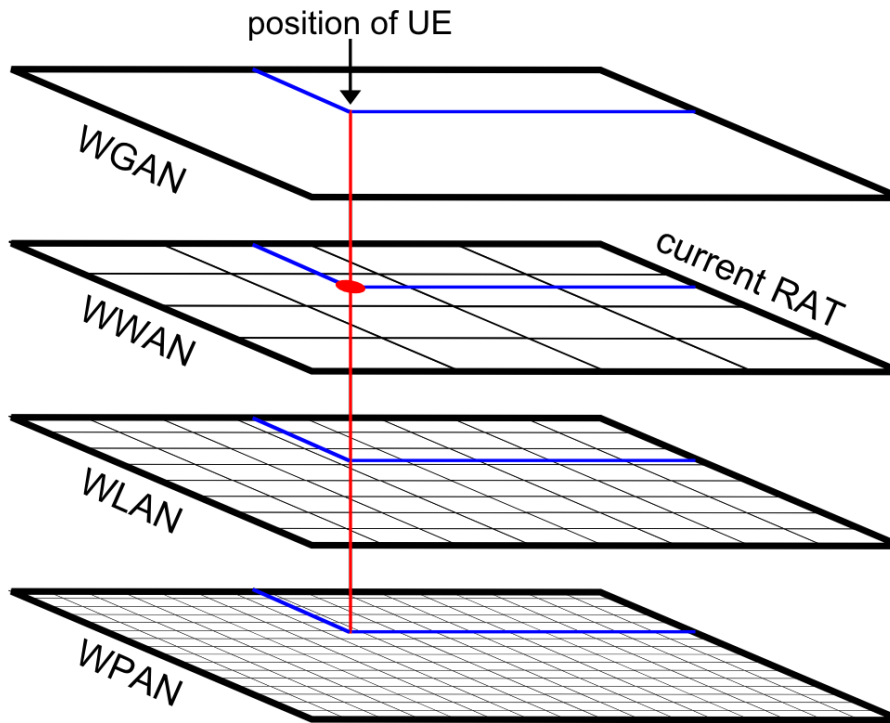


Clustering at time t_1



Clustering at time $t_1 + d$

Example: Inter-system Load Balancing



Force-based selection of target cell by each mobile

- Criteria/forces:
 - supported QoS in target cell (user view)
 - free capacity of target cell
 - cost per bit in target cell
 - migration penalty and handover cost

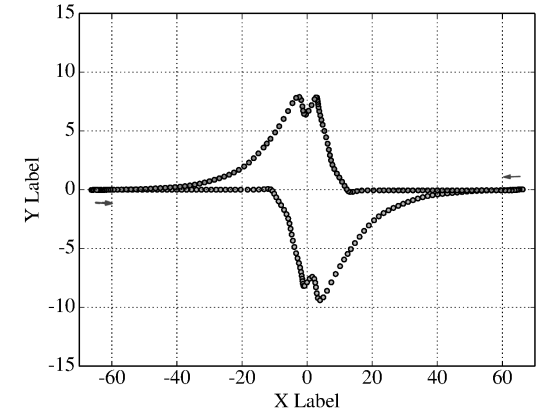
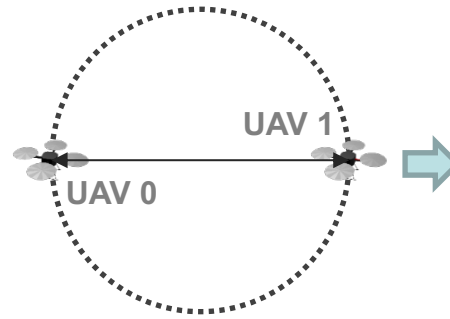
Source: A. Pillekeit, F. Derakhshan, E. Jugl, A. Mitschele-Thiel: "Force-based Load Balancing in Co-located UMTS/GSM Networks." 60th IEEE Vehicular Technology Conference (VTC 2004 Fall), 2004.

Example: Collision Avoidance for UAVs

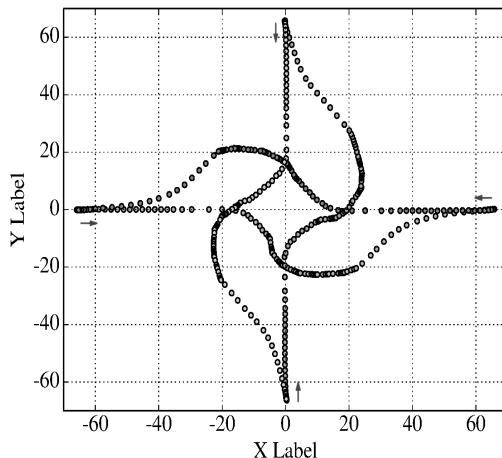
- Each UAV periodically broadcasts own coordinates and movement vector
- Receiving UAVs apply a force/torque model to adapt own movement

⇒ Emergence of roundabout

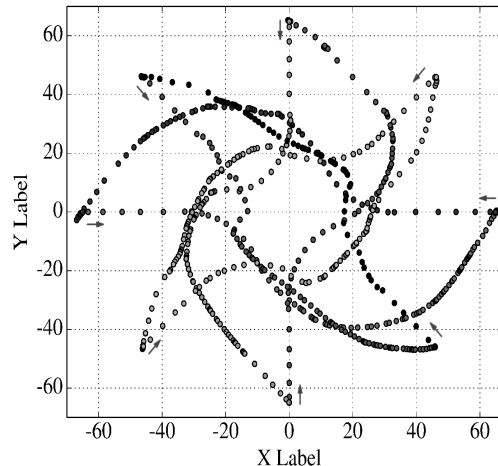
Scenario with 2 UAVs approaching each other



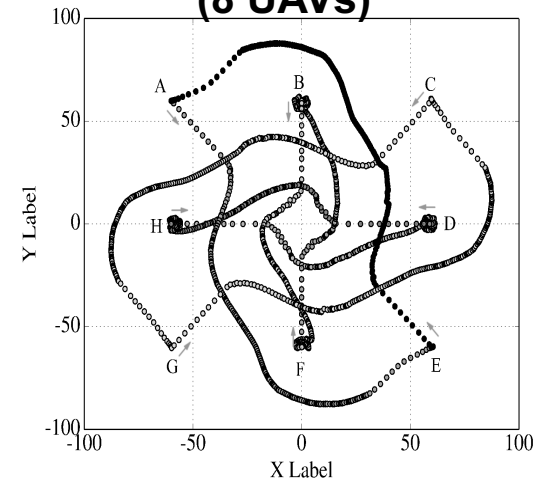
Roundabout with 4 UAVs



Roundabout with 8 UAVs



Double roundabout (8 UAVs)

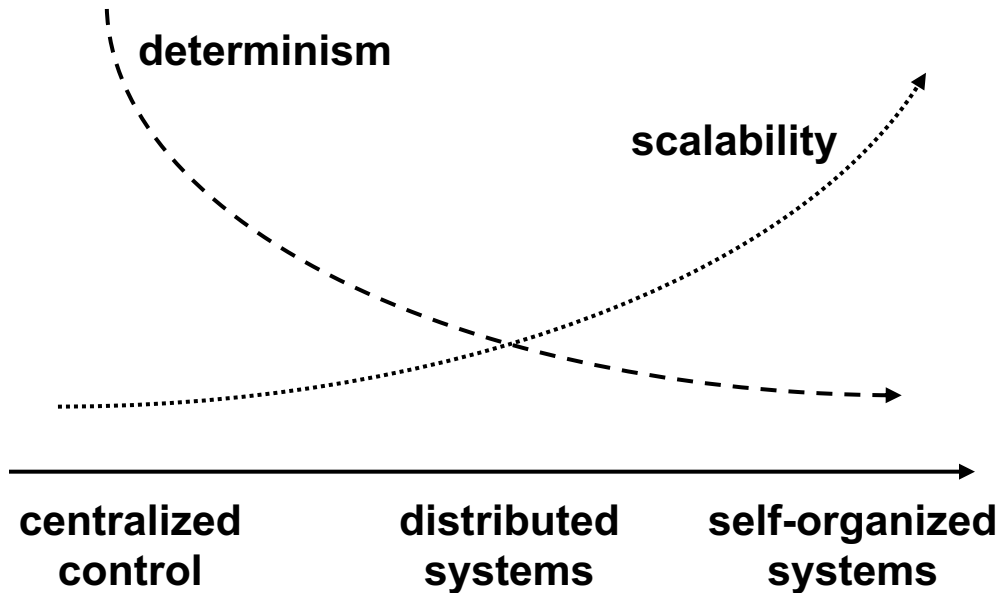


Challenges and Limits

Questions to be answered:

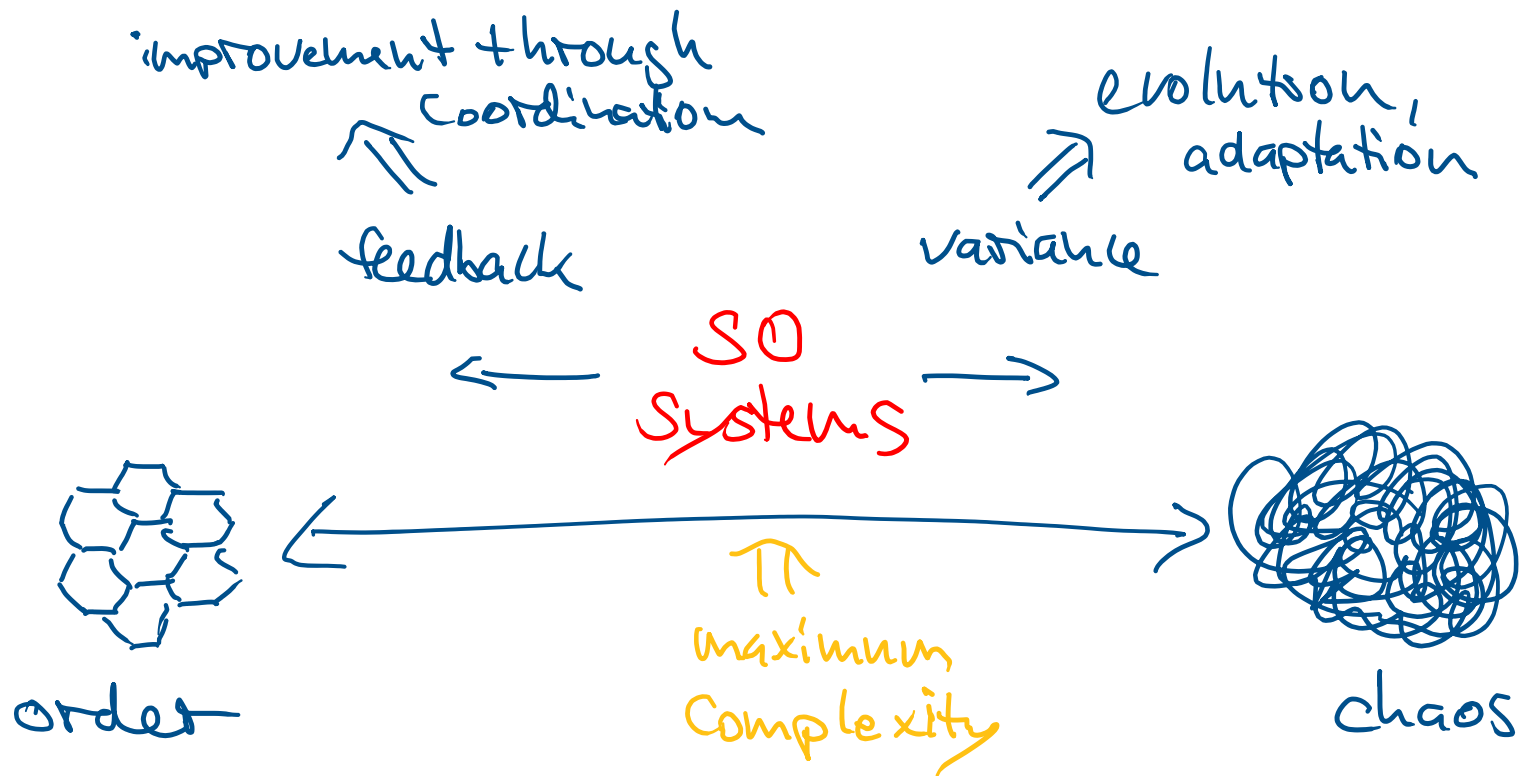
- What defines the Edge of Criticality? Why is it such an important issue?
- What are the advantages and disadvantages of operating close to this critical point? What are the properties of systems operating at that point?
- Identify the points of criticality for different applications as TCPs congestion control, RED, 802.11 DCF and the collision avoidance of UAVs
- Identify ways to reduce the risk of criticality for different SO algorithms
- Discuss how the term “overengineering” influences the risk of criticality.

Challenges and Limitations of Self-organization

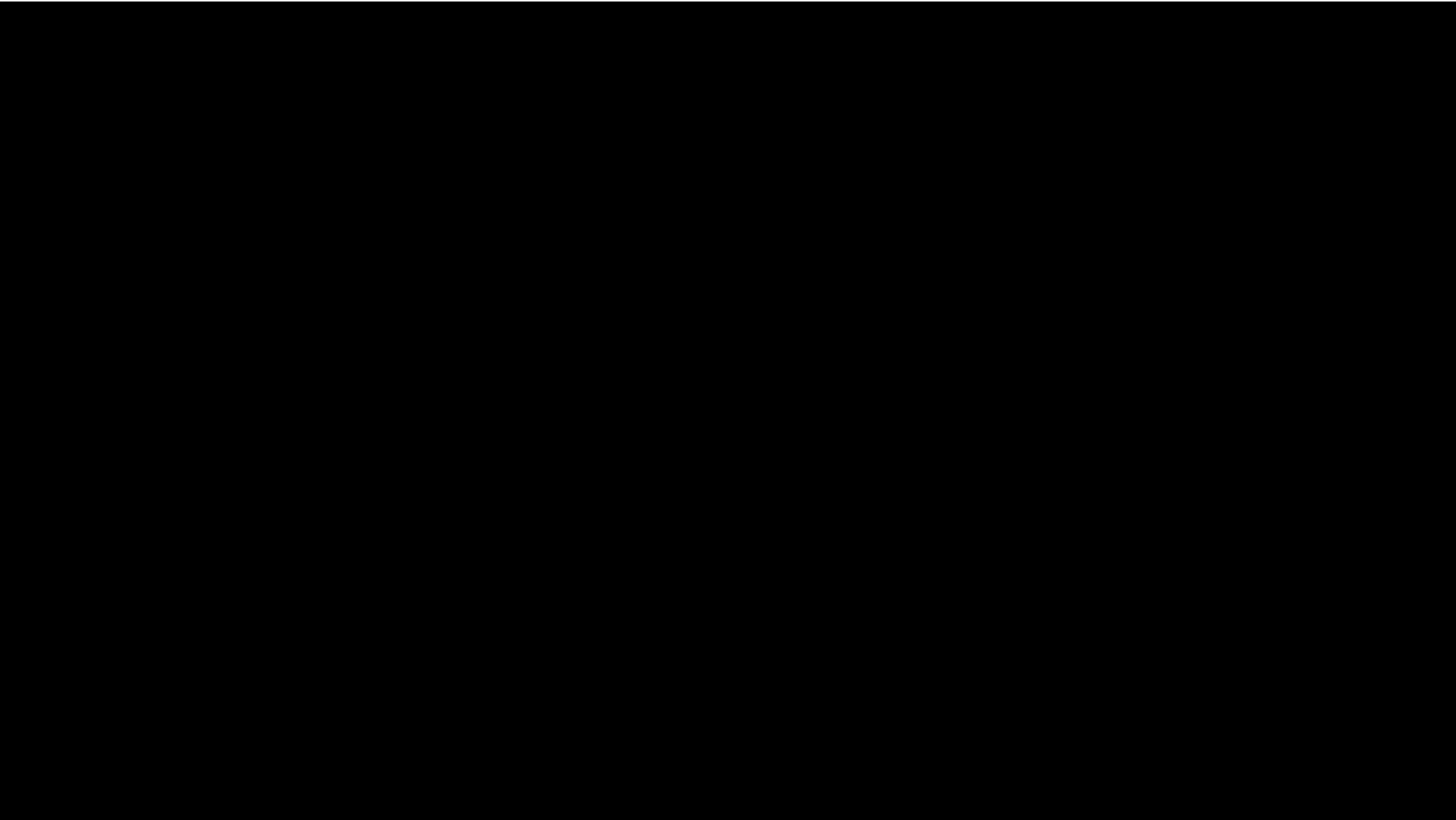


- Controllability
 - Predictability vs. scalability
- Software development
 - New software engineering approaches are needed
- System test
 - Handling of stochastic behavior
- Cross-mechanism interference
 - composition of multiple self-organizing mechanisms can lead to unforeseen effects

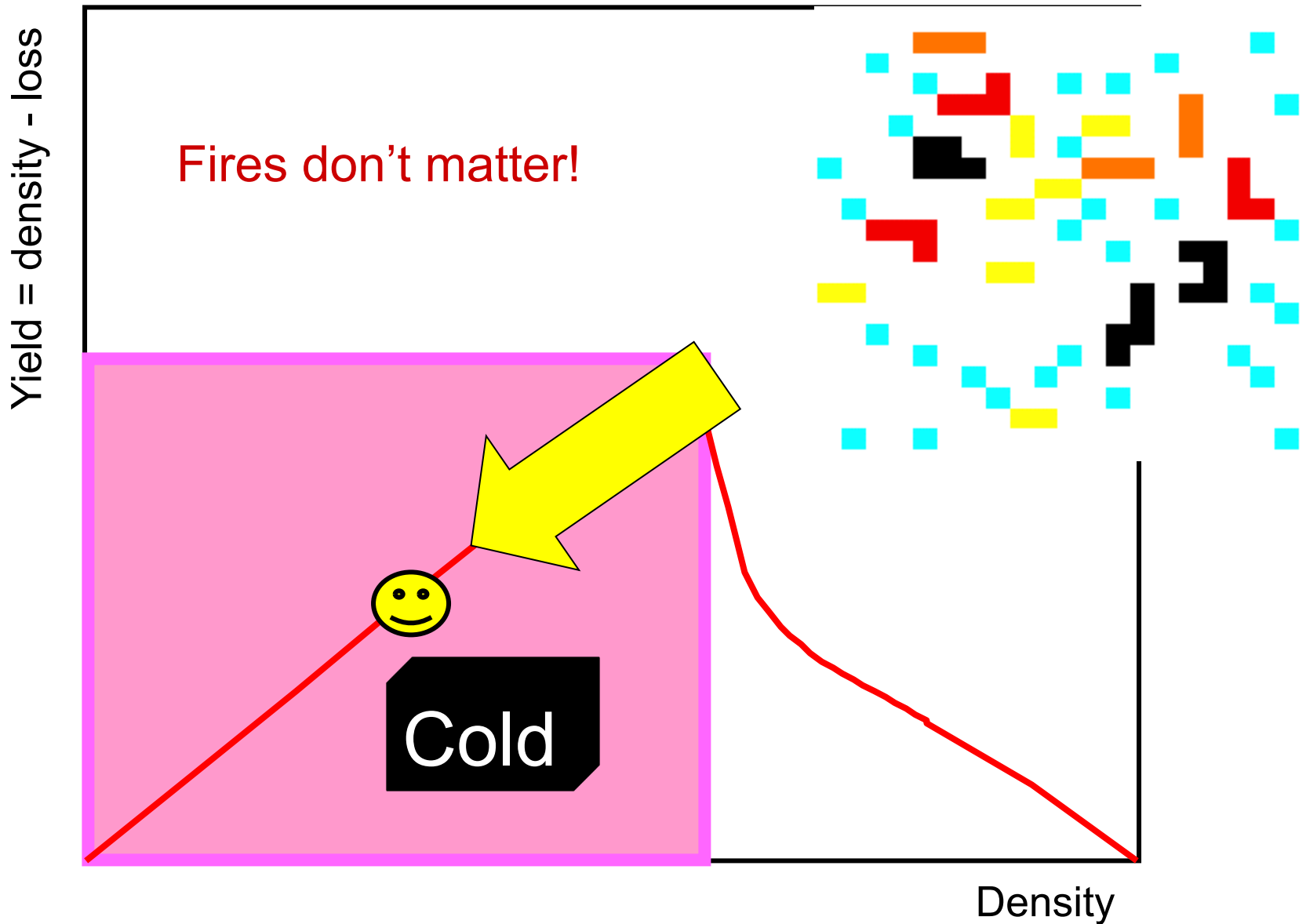
Order versus Chaos – The Art of SO System Design



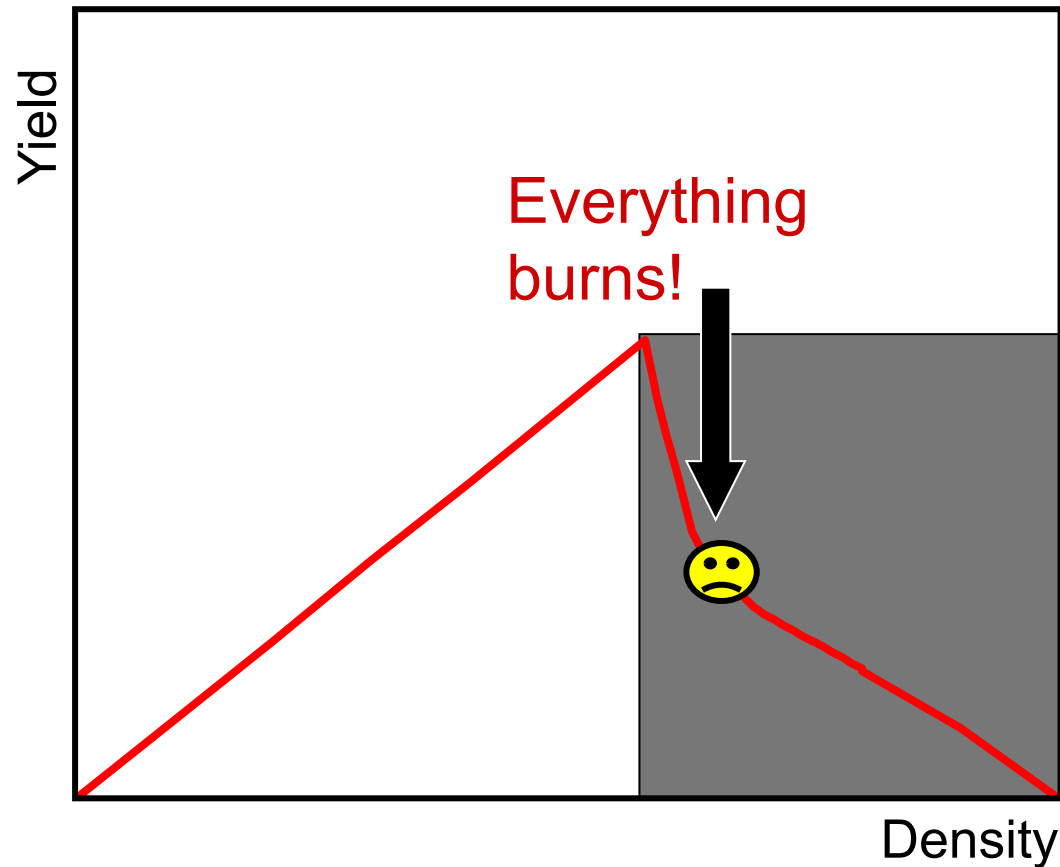
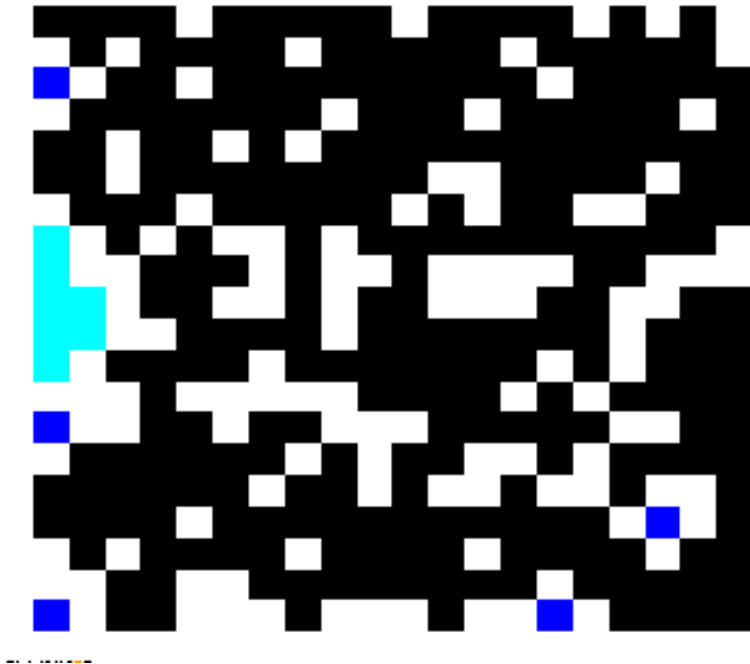
Robustness and resilience, self-organized criticality



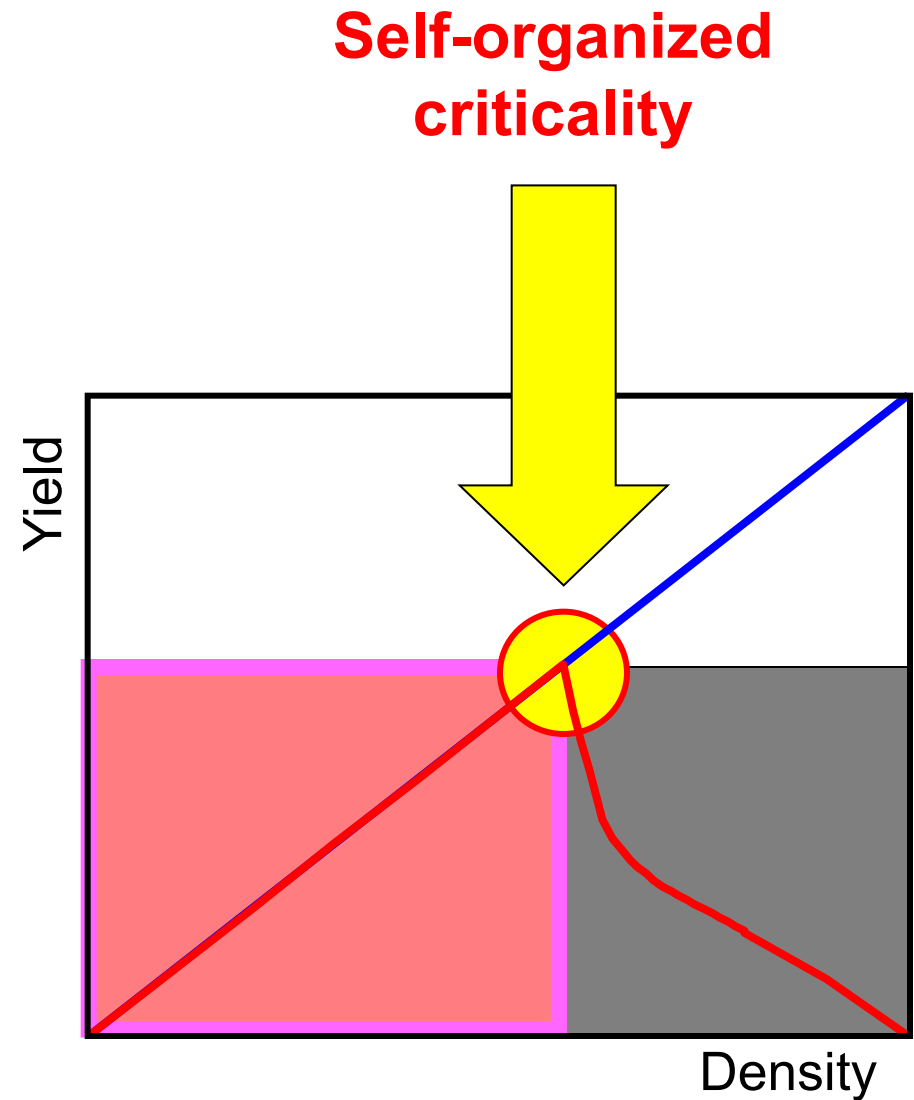
Problem: Stability and Predictability



Problem: Stability and Predictability



Self-organized criticality



The Edge of Chaos

The edge between

- order (i.e. coordination, patterns) and
- chaos (stochastics and variance)

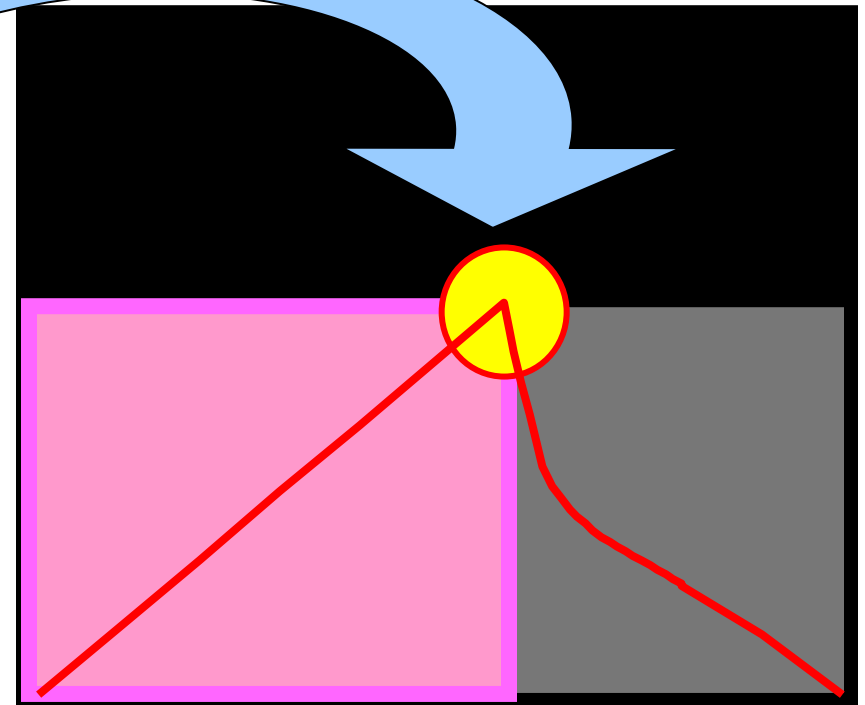
Point of operation where properties of the system change dramatically



Claims:

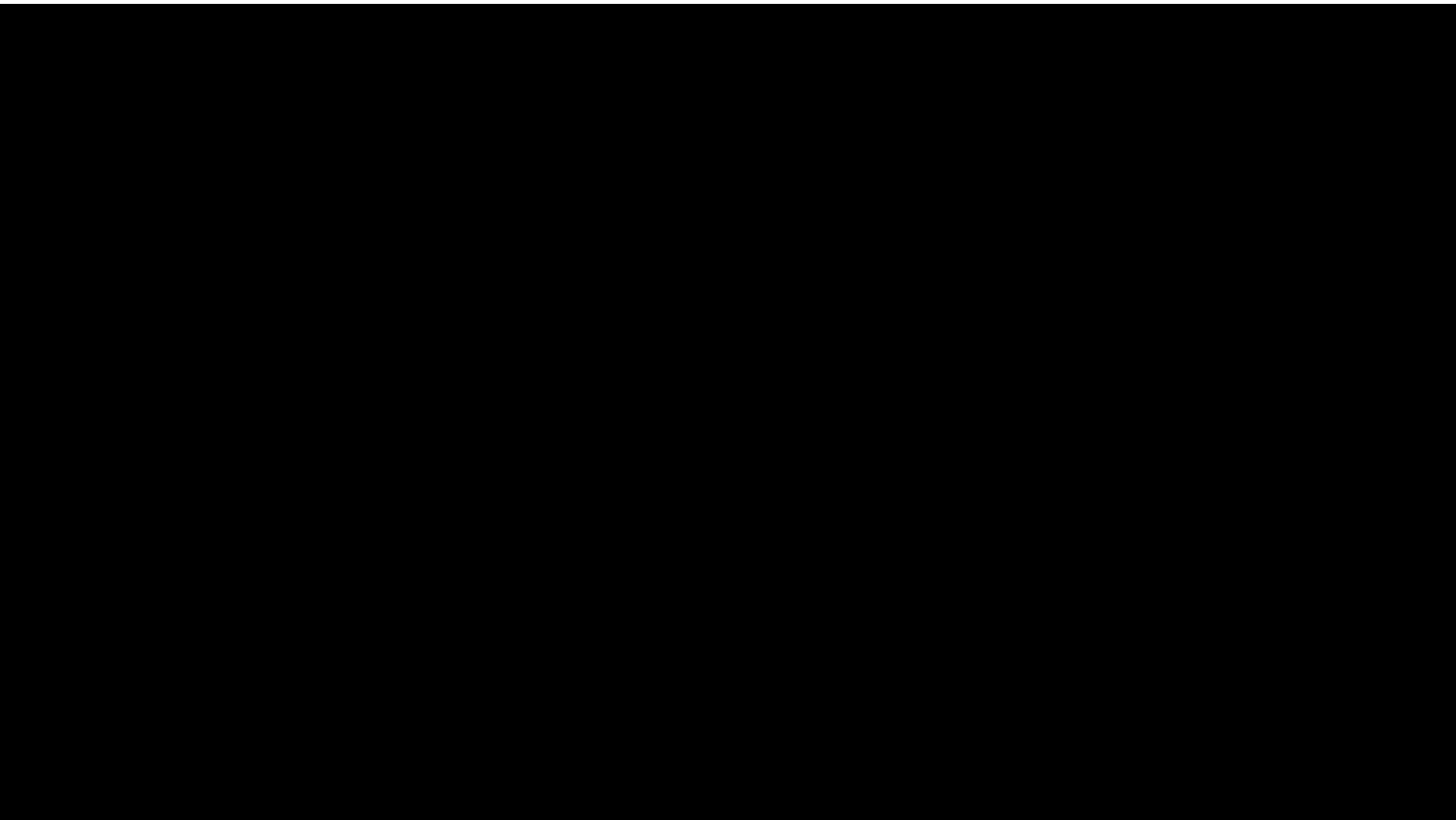
- Interesting phenomena is at criticality (or near a bifurcation)
- Life, networks, the brain, the universe operate best at the “edge of chaos”
- But, at the same time, are also extremely sensible to unforeseen changes

Yield



Density

Entropy and variety, edge of chaos



Conclusions

Self-organization: what does it mean?

- distributed, autonomous control of systems
- simple local algorithms or behavior
- interaction with neighbors and/or environment only

Nature

- provides a rich source of inspiration for computer systems
- has many features that are desirable for distributed computing such as auto-configuration, auto-organization, autonomy, scalability, flexibility, robustness, emergent behavior, and adaptability

Lots of applications of self-organization already

- self-organization keeps today's Internet together
- design and tuning of self-organized algorithms is highly complex
- stability is an issue

References

Introductory videos

- Introduction to self-organization (5:53)
https://www.youtube.com/watch?v=BTR17I_Eb_o
- Robustness and resilience, self-organized criticality (9:16)
<https://www.youtube.com/watch?v=HOTWIPmkdzo>
 - self-organized criticality (see 7:00)
- Entropy and variety, edge of chaos (8:47)
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 - edge of chaos (see 4:45)

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